

Laurent Valentin

Victoria, Seychelles

[GitHub](#) | [LinkedIn](#) | LaurentValentin@outlook.com

EDUCATION

University of Reading BSc (Hons) Meteorology and Climate, First Class Honours Top-scoring student in cohort.	Reading, United Kingdom 2021 – 2024
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Modules: Calculus, linear algebra, vector calculus, differential equations, statistics for weather and climate science, and numerical weather prediction.

Additional study: Audited CS3AI18 Artificial Intelligence, a final-year Computer Science module covering regression, classification, entropy, neural networks and reinforcement learning.

Additional training: Completed the three-course DeepLearning.AI *Mathematics for Machine Learning and Data Science Specialization* (2026), covering linear algebra, calculus, probability and statistics for machine learning.

AWARDS AND HONOURS

BSc Undergraduate Prize for Academic Achievement, University of Reading (2024)

RESEARCH EXPERIENCE

University of Reading <i>Undergraduate Researcher (UROP and Final-Year Dissertation)</i>	Reading, United Kingdom Jun 2023 – Apr 2024
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- **Dissertation (Sep 2023 – Apr 2024):** *Machine Learning-Based Sensitivity Analysis for Vegetation in Earth System Models*; supervised by Dr Ranjini Swaminathan and Dr Tristan Quaife.
- Leveraged JASMIN and Slurm to parallelise the training of 396 AdaBoost ensembles to compare how vegetation behaviour in UKESM1-0-LL and IPSL-CM6A-LR responded to precipitation, temperature and shortwave radiation across 44 IPCC regions.
- Demonstrated efficient sensitivity testing by comparing seasonal feature-importance rankings from Earth-system-model and observation-derived GPP emulators, revealing alignment and divergence with the observational benchmark.
- **UROP (Jun – Jul 2023):** *Evaluating GPP in the UK Earth System Model Using Artificial Intelligence*; supervised by Dr Ranjini Swaminathan. Applied the precursor ML workflow with UKESM1-0-LL and FLUXCOM data, including regional and seasonal preprocessing, model evaluation and feature-importance analysis.

OCEAN AI: Automatic Speech Recognition for Seychellois Creole <i>Researcher</i>	Seychelles 2026 – Present
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- Initiated a project targeting 100 hours of parliamentary speech after assessing the National Assembly's Hansard ASR need; identified limited domain data and trilingual code-switching as key constraints.
- Developed an FFmpeg, WhisperX and Meta MMS workflow to align National Assembly audio with normalised verbatim records and manually correct transcripts; working toward the first corrected hour for WER/CER baselines and later fine-tuning.

SELECTED PRESENTATIONS AND TECHNICAL REPORTS

- *Evaluating GPP in the UK Earth System Model Using Artificial Intelligence*: poster presentation at the University of Reading UROP Poster Showcase, Nov 2023.
- Technical feasibility assessment for a Seychellois Creole Hansard ASR system: report and presentation to the National Assembly of Seychelles, Jul 2026.
- Research presentation on AI initiatives in the Blue Economy in comparable small-island jurisdictions at the National Science, Technology and Innovation Forum, Seychelles, Jun 2026.

PROFESSIONAL EXPERIENCE

H. Savy Insurance

Seychelles

Business Analytics Engineer

Jan 2026 – Present

- As the company's sole engineer, built its first in-house analytics codebase using SQLAlchemy and Polars; the data layer has processed more than one million database rows for exploratory and operational data analysis.
- Established its first living technical and operational knowledge base, and diagnosed root causes of KYC data-quality issues affecting more than 10,000 customer records to support corrective action.

Seychelles Breweries Limited

Seychelles

Commercial Data Analyst (Data Engineering)

Jan 2025 – Dec 2025

- Standardised stock and depletion data from five distributors and about 1,200 outlets through a reusable Python ETL package, replacing preparation that could take days with a clean reporting dataset generated in under two minutes.
- Automated monthly payable-hours calculation for 60+ employees across six cost centres with a Python ETL and rules engine that paired raw biometric clock events into shifts and applied payroll rules at each overtime rate.

LEADERSHIP, TEACHING AND ENGAGEMENT

OCEAN AI

Seychelles

Founder and Chairperson

Dec 2025 – Present

- Founded and currently chairs a legally registered, six-person non-profit organisation focused on accessible AI and STEM education in Seychelles.
- Secured Absa Bank Seychelles sponsorship for a free nine-session beginner Python programme for 27 confirmed adult learners, and designed a game-based curriculum covering core Python concepts.
- Represented OCEAN AI as an invited panellist at the *U.S.-Seychelles Youth Dialogue & Innovation Panel Discussion*, discussing locally built technology, AI education and responsible adoption with approximately 60 participants.

H. Savy Insurance

Seychelles

Workplace Mentor

Mar 2026 – Aug 2026

- Designed and supervised two software projects: a Streamlit digital-KYC demo and a dependency scanner that parsed project dependencies, formatted packages for the OSV API and submitted HTTP requests; the scanner prompted the addition of vulnerability scanning to CI.
- Mentored an actuarial-science intern through a proof of concept for predicting first-year claim likelihood from customer and vehicle characteristics, covering exploratory analysis, feature selection, cross-validation, held-out evaluation and a final presentation.

TECHNICAL SKILLS

Programming: Python, SQL, Bash, L^AT_EX**Data engineering:** Polars, pandas, SQLAlchemy, Parquet**ML and scientific computing:** PyTorch, scikit-learn, XGBoost, AdaBoost, NumPy, xarray**Research methods:** Model evaluation, permutation importance, recursive feature elimination, speech data preparation**Tools and platforms:** Git, Linux/Unix, JASMIN, Slurm

ADDITIONAL UNIVERSITY ENGAGEMENT

- Mentor, University of Reading Internship Scheme, 2026 – Present.
- Former Student Ambassador and Campus Tours Team Lead, Student Course Representative and Race Equity Group Member, University of Reading.